

Instructions

- Answer all eight questions. Show sufficient working for method marks.
- A calculator may be used unless a question states otherwise.
- Write clearly in the spaces provided. Additional pages may be attached.

SCORE SAVED TO COURSE PROGRESS

38 / 42

1

Explain how natural selection can lead to antibiotic resistance in bacteria.

[6 marks]

Method sound; final simplification omitted.

6/6

2

Describe an experiment to investigate the effect of light intensity on the rate of photosynthesis.

[6 marks]

Check the condition before applying the rule.

6/6

AGENT ANNOTATED

3

Explain the difference between mitosis and meiosis and state where each occurs.

[5 marks]

Correct and clearly justified.

4/5

4

Use a Punnett square to predict the offspring ratio from two heterozygous parents.

[4 marks]

Method sound; final simplification omitted.

3/4

5

Describe how the body maintains a constant internal temperature.

[5 marks]

Check the condition before applying the rule.

5/5

AGENT ANNOTATED

6

Explain the role of the placenta in exchanging substances between mother and fetus.

[4 marks]

Correct and clearly justified.

3/4

7

Describe the structure of a cell membrane and explain how it controls what enters the cell.

[6 marks]

Method sound; final simplification omitted.

6/6

8

Explain the difference between diffusion, osmosis, and active transport, giving one example of each.

[6 marks]

Check the condition before applying the rule.

5/6